PROCEDURE: Real-time bedside ultrasound-guided diagnostic and therapeutic thoracentesis.

INDICATION: Symptomatic, large, recurrent right-sided pleural effusion in a 69-year-old male with decompensated systolic heart failure, producing respiratory distress and hypoxemia.

PROCEDURE IN DETAIL: The patient was seated upright and leaned forward across an adjustable bedside table supported by pillows. Thoracic ultrasound was performed utilizing a 3.5 MHz curvilinear probe, which confirmed a large, anechoic, free-flowing right pleural fluid collection without significant septations or loculations. An optimal puncture site was identified in the posterior axillary line at the 8th intercostal space, well above the diaphragm. The skin was marked, widely prepped with 2% chlorhexidine gluconate in 70% isopropyl alcohol, and draped with sterile barrier towels.

Local anesthesia was established with 1% plain lidocaine, infiltrating the epidermis, subcutaneous tissues, and the superior margin of the 9th rib to avoid the intercostal neurovascular bundle. The parietal pleura was infiltrated, and 5 mL of clear straw-colored fluid was aspirated confirming entry into the pleural space. A 5-French catheter-over-needle thoracentesis device was introduced under negative pressure until fluid was aspirated, at which point the over-the-needle radiopaque polyurethane catheter was smoothly advanced into the pleural space and the introducing needle withdrawn.

A three-way stopcock was attached to the catheter. Diagnostic fluid aliquots were collected in sterile vacutainers for analysis. Therapeutic drainage was initiated via negative pressure vacuum canisters. A total of 1,250 mL of amber serous pleural fluid was evacuated over 25 minutes. Fluid drainage was discontinued when the patient developed mild sub-sternal chest tightness and persistent coughing. The catheter was quickly removed at end-expiration, and a sterile occlusive pressure dressing was applied.

LABORATORY RESULTS: Pleural fluid total protein 2.1 g/dL (serum protein 6.4 g/dL, ratio 0.33), fluid LDH 84 U/L (serum LDH 220 U/L, ratio 0.38), total nucleated cell count 420/uL with 88% lymphocytes. These parameters satisfy Light's criteria for a transudative pleural effusion secondary to congestive heart failure. Post-procedure upright chest radiograph demonstrated significant re-expansion of the right lower lobe without evidence of iatrogenic pneumothorax or hemothorax.